

Lesson 64 Practice: Current Cybersecurity, Cryptography and Cyber Law

JKSSB Computer Awareness

October 10, 2026

Provenance: All 25 questions are original exam-pattern questions (*[Original]* plus the *[JKSSB FAA-pattern]*, *[JKPSI-pattern]*, *[SSC CGL-pattern]*, and *[JKSSB-pattern]* extension items, including MSQs); none reproduces an official paper item.

Q1. *[Original]* A parasitic (file-infector) virus is best described as:

- A. a virus that attaches itself to an executable file and runs when the host program runs
- B. a worm that spreads across a network without any host program
- C. a Trojan horse that disguises itself as useful software without self-replicating
- D. a flood of traffic from many machines that overwhelms a server

Q2. *[Original]* Expand DES.

- A. Data Encryption Standard
- B. Digital Encryption System
- C. Distributed Encoding Scheme
- D. Data Evaluation Service

Q3. *[Original]* DES processes data in blocks of:

- A. 32 bits
- B. 56 bits
- C. 64 bits
- D. 128 bits

Q4. *[Original]* Which of the following is **NOT** true of a digital signature?

- A. It proves the sender's authenticity.
- B. It proves the message was not altered in transit.
- C. It provides integrity of the message.
- D. It encrypts the message body so only the recipient can read it.

Q5. *[Original]* Encryption primarily provides _____, whereas a digital signature primarily provides _____.

- A. confidentiality; authenticity and integrity

- B. authenticity; confidentiality
 - C. availability; confidentiality
 - D. integrity; availability
- Q6.** *[Original]* Which firewall type examines only per-packet headers such as source/destination address, port, and protocol?
- A. Packet-filtering firewall
 - B. Stateful-inspection firewall
 - C. Application (proxy) firewall
 - D. Personal anti-virus program
- Q7.** *[Original]* A firewall that remembers open sessions and drops packets that do not belong to a known session is a:
- A. packet-filtering firewall
 - B. stateful-inspection firewall
 - C. boot-sector virus
 - D. Trojan horse
- Q8.** *[Original]* Which firewall acts as a go-between, inspecting application-layer content and commands?
- A. Packet-filtering firewall
 - B. Stateful-inspection firewall
 - C. Application (proxy) firewall
 - D. Spam filter of the mail client only
- Q9.** *[Original]* A fraudulent mail styled as a bank KYC notice asks a clerk for her OTP on a fake login page. This is:
- A. phishing
 - B. ransomware
 - C. DDoS
 - D. disk defragmentation
- Q10.** *[Original]* A clerk's files are suddenly unreadable, with a message demanding payment for the decryption key. This is:
- A. phishing
 - B. ransomware
 - C. DDoS
 - D. hacking without any malicious goal
- Q11.** *[Original]* Which of the following correctly distinguishes the four threats?
- A. Phishing floods a server; ransomware tricks the user; DDoS encrypts files; hacking defragments disks.

- B. Phishing tricks the user into revealing credentials; ransomware locks files for ransom; DDoS floods a service; hacking is unauthorised access generally.
- C. Phishing encrypts files; ransomware floods a server; DDoS reveals OTPs; hacking is a firewall type.
- D. Phishing is a firewall; ransomware is an anti-virus; DDoS is encryption; hacking is a backup.

Q12. *[Original]* Expand DDoS.

- A. Distributed Denial of Service
- B. Direct Disk Operating System
- C. Digital Data on Server
- D. Dynamic Domain of Service

Q13. *[Original]* Evaluate the statements:

- I. DES is a symmetric cipher: the same key encrypts and decrypts.
 - II. DES works on 64-bit blocks of data.
 - III. A digital signature by itself keeps the message body secret from everyone except the recipient.
- A. I only
 - B. I and II only
 - C. II and III only
 - D. I, II, and III

Q14. *[Original]* Which pairing of firewall type and function is correctly matched?

- A. Packet-filtering — inspects application content as a go-between
- B. Stateful inspection — tracks connection state in addition to headers
- C. Application (proxy) — decides each packet by headers only
- D. Packet-filtering — disinfects infected executable files

Q15. *[Original]* Which of the following is a governance concern of the Internet at awareness level?

- A. Which country's law applies when an act crosses borders
- B. The colour scheme of a single office website
- C. The brand of printer attached to the office desktop
- D. The seating plan of the computer laboratory

Q16. *[Original]* Which data-protection principle requires that personal data be used only for the purpose stated at collection?

- A. Purpose limitation (stated-purpose use)
- B. Unlimited retention
- C. Open sharing without consent

- D. Default publication
- Q17.** *[Original]* Copying licensed software without permission is, at awareness level, an example of:
- A. copyright infringement
 - B. fair dealing for private study
 - C. first ownership
 - D. lawful backup under licence
- Q18.** *[Original]* Match the copyright concept to what it covers: (a) meaning of copyright, (b) first ownership, (c) infringement, (d) fair dealing. Which match is correct?
- A. (a)—punishment for wilful violation; (b)—permitted study uses; (c)—who owns first; (d)—owner's exclusive rights
 - B. (a)—owner's exclusive rights; (b)—who owns the work first; (c)—what counts as violation; (d)—limited permitted uses such as study
 - C. (a)—who owns first; (b)—owner's rights; (c)—permitted uses; (d)—what counts as violation
 - D. (a)—permitted uses; (b)—violation; (c)—owner's rights; (d)—first ownership
- Q19.** *[Original]* Expand ICANN.
- A. Internet Corporation for Assigned Names and Numbers
 - B. International Computer and Network Node
 - C. Internet Code for Access and Navigation
 - D. Indian Council for Assigned Network Names
- Q20.** *[Original]* An office server slows to a halt under a flood of traffic from many machines, so genuine users cannot reach it. The correct response pair is:
- A. identify it as DDoS and filter/block the flood at the network barrier (firewall/ISP level)
 - B. identify it as phishing and ask users to change their OTPs only
 - C. identify it as a file-infector virus and reinstall the printer driver
 - D. identify it as fair dealing and take no technical action
- Q21.** *[JKSSB FAA-pattern]* Encryption is used to provide:
- A. confidentiality of data.
 - B. louder speaker output.
 - C. brighter monitor light.
 - D. faster paper printing.
- Q22.** *[JKPSI-pattern]* A digital signature provides which guarantees?
- A. Authenticity and integrity
 - B. Free printer ink
 - C. Louder ringtones

- D. Heavier paper
- Q23.** *[SSC CGL-pattern]* Which law deals with cybercrime in India?
- A. The Information Technology Act
 - B. The Paper Weight Act
 - C. The Speaker Volume Act
 - D. The Ink Colour Act
- Q24.** *[JKSSB-pattern]* (MSQ: select ALL correct options.) Which are symmetric-key concepts or examples?
- A. Same key encrypts and decrypts
 - B. AES cipher
 - C. Fast bulk-data encryption
 - D. Public-private key pairs
- Q25.** *[SSC CGL-pattern]* (MSQ: select ALL correct options.) Which statements about cryptography are correct?
- A. Encryption hides message meaning.
 - B. Decryption restores the original message.
 - C. Hashing checks data integrity.
 - D. Encryption makes data print faster.
- Q26.** *[PYQ-adapted: SSC CGL 2025 Tier-II Paper-I, 19-Jan-2026 Q5; SRC-18]* Which option provides message-level email encryption to protect content from unauthorized access?
- A. S/MIME
 - B. IMAP
 - C. SSL alone
 - D. SMTP
- Q27.** *[PYQ-adapted: SSC CGL 2025 Tier-II Paper-I, 19-Jan-2026 Q11; SRC-18]* What is a common purpose of a Trojan horse?
- A. Disguise malicious activity as a benign program
 - B. Self-replicate as a worm
 - C. Filter network packets as a firewall
 - D. Repair operating-system vulnerabilities
- Q28.** *[Original]* Which property does encryption primarily provide when only authorized parties can read the content?
- A. Confidentiality
 - B. Availability
 - C. Non-repudiation only

D. Compression

Q29. *[Original]* A digital signature primarily helps verify:

- A. Authenticity and integrity
- B. Secrecy of the message content by itself
- C. Internet connection speed
- D. File size reduction

Q30. *[Original, MSQ drill]* (MSQ: select ALL correct options.) Match security concept to purpose.

- A. Encryption — confidentiality
- B. Digital signature — authenticity/integrity
- C. Firewall — traffic control
- D. Antivirus — guaranteed prevention of every attack